Adaptive Cyber-Physical Security Phase-1 Project Report

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***Abstract*—This report presents an adaptive intrusion detection pipeline applied to the CIC-IDS-2018 network-traffic benchmark [1], with an emphasis on *zero-day* threat recognition. The processed dataset contains 2,097,149 labeled flows and spans four traffic classes. Two complementary models are studied: a supervised Random Forest [3] configured under a seen-versus-unseen attack split, and a One-Class SVM [4] trained exclusively on benign traffic. The Random Forest achieved 0.4396 accuracy, 1.00 precision, 0.178 recall, 0.303 F1, and 0.00 zero-day recall; the One-Class SVM achieved 0.907 precision, 0.145 recall, 0.249 F1, and 0.145 zero-day recall. The complementary nature of the two models—high confidence on known threats versus novelty sensitivity for new ones—demonstrates the necessity of a hybrid evaluation strategy for adaptive cyber-physical security [6].**

***Index Terms*—Intrusion Detection, CIC-IDS-2018, Random Forest, One-Class SVM, Zero-Day Detection, Anomaly Detection, Cyber-Physical Security, Feature Engineering**

I. Introduction and Problem Statement

Cyber-physical systems (CPS) integrate networked computation with physical processes, making them high-value targets for sophisticated adversaries. Intrusion detection in such environments must handle two fundamentally different conditions: (i) classifying attacks that appear in historical training data, and (ii) flagging novel attack traffic not present during training—the *zero-day* scenario. Relying on purely supervised classifiers addresses the first condition but systematically fails the second [2], because attack taxonomies evolve faster than labeled corpora. Guo's comprehensive review of ML-based zero-day detection confirms that no single model achieves uniform accuracy across all attack families [6].

This project addresses both conditions by pairing a discriminative classifier with an anomaly detector applied to CIC-IDS-2018 network flow data [1]. Complementary benchmark datasets such as CIC-IDS-2017 [7] and NSL-KDD [8] informed design decisions around feature engineering and evaluation protocol. The practical pipeline:

- cleans and standardizes raw traffic features;
- studies class balance and correlation structure;
- reduces redundant and unstable features;
- evaluates a supervised baseline for known attacks; and
- evaluates an anomaly model for zero-day detection.

II. Literature Review

CIC-IDS-2018. Sharafaldin et al. introduced CIC-IDS-2018 to address benchmark limitations such as outdated attack patterns, weak traffic diversity, and unrealistic feature representations [1]. The dataset provides modern network flow statistics with explicit attack labels, making it well-suited for both supervised and anomaly-based evaluation. This project uses CIC-IDS-2018 as its primary benchmark [1].

CIC-IDS-2017 and NSL-KDD. Prior to CIC-IDS-2018, the CIC-IDS-2017 dataset [7] established a widely-used five-day traffic capture covering brute-force, DoS, DDoS, and botnet attacks. The NSL-KDD dataset [8] remains a foundational benchmark with 41 network features across four attack categories: DoS, probing, Remote-to-Local (R2L), and User-to-Root (U2R). Both datasets informed the feature engineering and evaluation strategies adopted in this work.

Zero-Day Attack Detection. Guo's survey [6] provides a comprehensive review of ML-based zero-day attack detection, covering unsupervised outlier methods, supervised and hybrid learning, and transfer learning approaches. The survey identifies a fundamental challenge: zero-day attack samples are unavailable at training time by definition, forcing models to generalise from known attack distributions. One-Class SVM and autoencoder-based detectors are highlighted as the dominant unsupervised approaches, with recall varying substantially across attack types—a finding directly reflected in the results of this project.

Random Forest. Breiman's ensemble formulation reduces prediction variance through bagging and random feature subsampling [3]. In intrusion detection, this robustness is valuable because packet-derived features exhibit collinearity, noise, and heterogeneous scales. Recent surveys confirm Random Forest remains a strong tabular baseline for network security tasks [2].

One-Class SVM. Schölkopf et al. formalized One-Class SVM as a support estimation problem [4]. This framing fits zero-day detection naturally: instead of requiring exhaustive malicious labels, the model estimates the boundary of normal traffic and flags deviating samples. Guo [6] evaluates One-Class SVM against autoencoder methods on both CIC-IDS-2017 [7] and NSL-KDD [8], confirming that One-Class SVM achieves

TABLE I
PROCESSED DATASET CLASS DISTRIBUTION

Class / Attack Type	Count
Benign	1,209,156
DoS attacks-Hulk	461,912
Bot	286,191
DoS attacks-SlowHTTPTest	139,890
Total	2,097,149

high precision on recognisable deviations but struggles with attacks that closely resemble benign traffic.

Anomaly Detection. Chandola et al. survey anomaly detection as a family of techniques built on diverse assumptions about what "normal" looks like [5]. Their taxonomy clarifies why both supervised and anomaly-based detectors contribute complementary information—the supervised model excels when representative attack labels exist, while the anomaly detector hedges against label poverty.

Network Security Challenges. Buczak and Guven highlight recurring ML challenges in cyber-security: label quality, class imbalance, concept drift, and feature redundancy [2]. These challenges are directly observable in the CIC-IDS-2018 [1], CIC-IDS-2017 [7], and NSL-KDD [8] benchmarks, all of which exhibit skewed class distributions and correlated feature spaces that motivate the preprocessing steps described in Section IV.

III. Dataset Quality and Exploratory Analysis

A. Dataset Snapshot

The processed dataset used in this project contains 2,097,149 rows and 52 columns after cleaning, derived from the CIC-IDS-2018 benchmark [1]. Table I summarises the label distribution.

The dataset exhibits strong class imbalance: benign traffic accounts for $\approx 57.7\%$ of samples, while the three attack categories are unevenly distributed among themselves. This imbalance directly motivates the use of precision and recall as primary metrics rather than raw accuracy, consistent with evaluation practice on NSL-KDD [8] and CIC-IDS-2017 [7].

B. Quality Observations

- Benign traffic dominates the dataset; attack categories are unevenly distributed.
- Raw network features contain infinities, missing values, and identifier columns unsuitable for learning.
- Many derived traffic features are highly correlated, inflating compute cost and distorting model interpretation.
- A stray header-like row labeled `label` appears in the processed artifact and must be removed explicitly.

C. EDA Visualisations

Figure 1 illustrates the strong class skew. Figures 2 and 3 reveal heavy-tailed feature distributions and substantial correlation across packet-length, inter-arrival-time, and segment statistics—patterns that directly motivate feature pruning and scale-aware modelling [2].

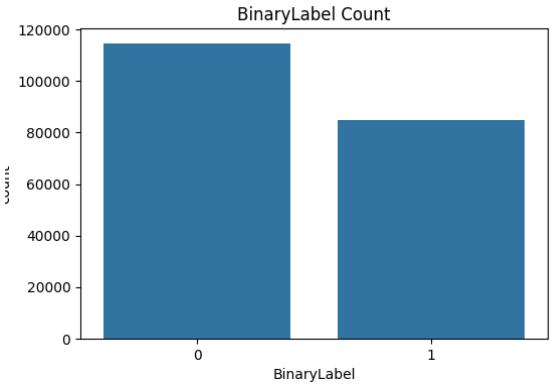

Fig. 1. Class distribution in the processed CIC-IDS-2018 dataset [1].

Fig. 2. Representative feature histograms showing heavy-tailed distributions.

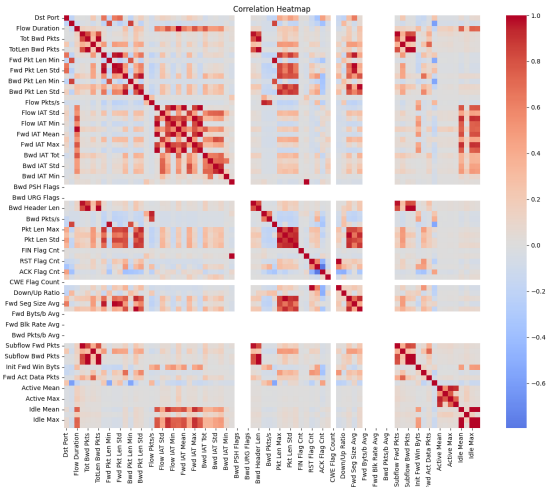

Fig. 3. Correlation structure before feature reduction reveals dense collinearity blocks.

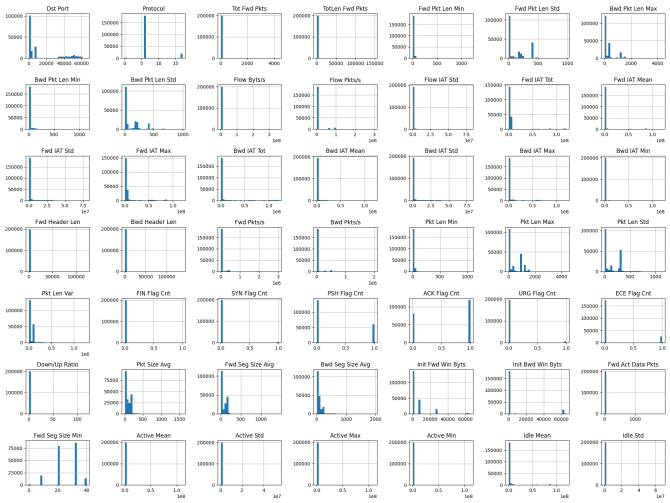

Fig. 4. Feature distributions after cleaning and correlation-based reduction.

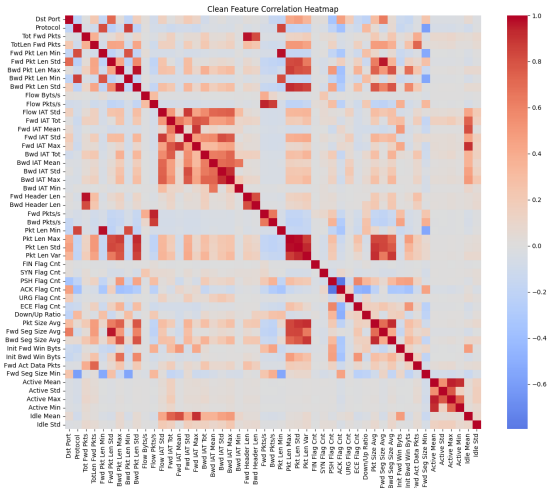

Fig. 5. Correlation heatmap after feature reduction; collinearity blocks are substantially reduced.

IV. Feature Engineering

The feature-engineering pipeline follows four main phases:

1) **Identifier removal.** Drop `Flow ID`, source/ destination IPs, port identifiers, and timestamps, which leak non-general information.
2) **Numeric coercion.** Coerce all non-label columns to numeric types; replace infinite values with `NaN` and impute or drop sparse features.
3) **Constant removal.** Discard features with zero variance, as they carry no discriminative information.
4) **Correlation pruning.** Apply a Pearson correlation threshold of $\rho = 0.98$: retain domain-preferred features and drop redundant partners. This reduces the feature set from 82 raw columns to 52 meaningful columns, consistent with preprocessing practice reported for CIC-IDS-2017 [7] and NSL-KDD [8].

Figures 4 and 5 confirm that the cleaned feature set has a more compact correlation structure compared to the raw features, supporting both model interpretability and training efficiency [2].

V. Theoretical Background

A. Random Forest

A Random Forest trains B decision trees $\{T_b\}_{b=1}^{B}$, each on a bootstrap sample of the training data with a random subset of $m \ll p$ features at every split [3]. Classification prediction is:

$$\hat{y}(\mathbf{x}) = \arg\max_{c} \sum_{b=1}^{B} \mathbf{1}[T_b(\mathbf{x}) = c]\,. \tag{1}$$

Bagging reduces variance without increasing bias; feature subsampling de-correlates trees and further lowers the ensemble variance [3]. For network-flow tabular data with many correlated numeric predictors, Random Forest provides a strong baseline that does not require feature scaling.

B. One-Class SVM

One-Class SVM estimates a decision function $f(\mathbf{x}) = \operatorname{sgn}\big(w^\top \phi(\mathbf{x}) - \rho\big)$ by solving [4]:

$$\min_{w,\,\rho,\,\xi} \; \frac{1}{2}\|w\|^2 + \frac{1}{\nu n}\sum_{i=1}^{n} \xi_i - \rho \tag{2}$$

subject to

$$w^\top \phi(\mathbf{x}_i) \geq \rho - \xi_i, \qquad \xi_i \geq 0, \tag{3}$$

where $\nu \in (0, 1]$ upper-bounds the expected outlier fraction and $\phi(\cdot)$ is the kernel mapping [4]. An RBF kernel $k(\mathbf{x}, \mathbf{x}') = \exp\big(-\gamma\|\mathbf{x} - \mathbf{x}'\|^2\big)$ is used here, which provides implicit infinite-dimensional feature maps without explicit computation. After standardisation ($\mu = 0$, $\sigma = 1$), the model is trained on benign traffic only and evaluated on both benign and all attack samples, treating every attack as a zero-day instance [6].

C. Evaluation Metrics

Two evaluation perspectives are maintained throughout:

- **Standard supervised metrics:** accuracy, precision, recall, and F1-score, measured on the full test set.
- **Zero-day recall:** recall computed *exclusively* on the unseen-attack subset [6], defined as

$$\text{ZD-Recall} = \frac{TP_{\text{unseen}}}{TP_{\text{unseen}} + FN_{\text{unseen}}}. \tag{4}$$

This separation is critical: a model can score high aggregate metrics while still exhibiting zero zero-day recall—failing the core security requirement of novelty detection [6].

VI. Model Application and Results

A. Random Forest — Seen/Unseen Split

The training regime follows a seen-versus-unseen split on CIC-IDS-2018 [1]:

TABLE II
RANDOM FOREST METRICS (RECORDED REPOSITORY RUN)

Metric	Value
Accuracy	0.4396
Precision	1.0000
Recall	0.1782
F1-score	0.3025
Zero-Day Recall	0.0000

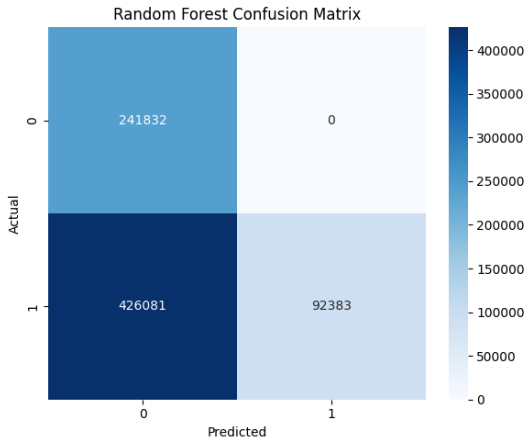

Fig. 6. Random Forest confusion matrix. High precision but zero zero-day recall confirms over-conservatism on unseen attacks [6].

TABLE III
ONE-CLASS SVM METRICS (RECORDED REPOSITORY RUN)

Metric	Value
Precision	0.9073
Recall	0.1445
F1-score	0.2493
Zero-Day Recall	0.1445

- benign traffic: split 80/20 train/test;
- `DoS attacks-Hulk`: treated as a *seen* attack included in training;
- `Bot` and `DoS attacks-SlowHTTPTest`: withheld as *unseen* attacks to evaluate zero-day recall.

Table II reports the recorded metrics, and Figure 6 shows the corresponding confusion matrix.

B. One-Class SVM — Anomaly Configuration

The One-Class SVM is trained on benign traffic only, evaluated on held-out benign samples plus *all* attack classes—making every attack effectively zero-day [6]. Standard scaling (z-score) is applied before fitting the RBF-kernel model [4]. Table III and Figure 7 summarise the results.

C. Interpretation

The two models serve complementary operational roles, consistent with the hybrid strategy advocated by Guo [6] and Chandola et al. [5]:

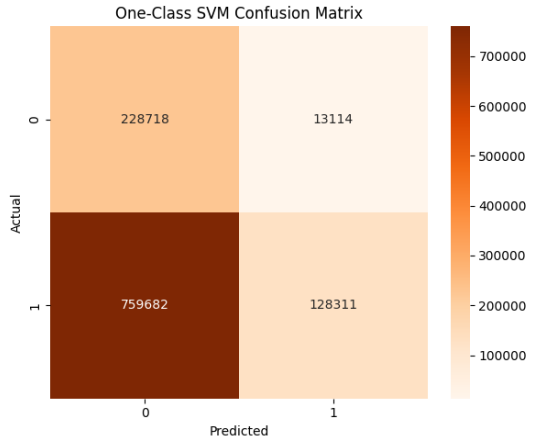

Fig. 7. One-Class SVM confusion matrix. Non-zero zero-day recall means the model detects some novel attacks; low recall reflects the inherent conservatism of support estimation [4].

- **Random Forest** [3] achieves perfect precision on training-distribution attacks but *zero* zero-day recall.
- **One-Class SVM** [4] achieves lower overall recall but *non-zero* zero-day recall, providing a hedge against novel attack families.

This confirms the central thesis: a purely supervised classifier is insufficient when the evaluation target includes unseen attacks [2]. A hybrid pipeline combining both models is therefore more defensible—the supervised component provides high-confidence classification for known threats, while the anomaly detector acts as a novelty sentinel [6].

VII. REPOSITORY AND CODE QUALITY

The repository mirrors a standard data-science workflow:

- `pipeline/` — preprocessing and cleaning scripts;
- `experiments/eda` — exploratory visualisations;
- `experiments/feature_engineering` — correlation and histogram analysis;
- `experiments/models` — training and evaluation scripts;
- `outputs/plots` — auto-generated figures; and
- `report/` — written deliverables.

Recent code-quality improvements include: root-relative path resolution replacing fragile working-directory assumptions; explicit failure on missing required inputs; safe removal of the stray `label` row during preprocessing; crash-hardening for small benign subsets in the One-Class SVM script; and cache-path hardening for reproducibility in restricted environments. The primary remaining engineering gap is a top-level orchestrated runner with lightweight schema validation tests.

VIII. LIMITATIONS AND NEXT STEPS

- The checked-in artifact covers a subset of the full CIC-IDS-2018 [1] attack space; extended evaluation requires regeneration from the complete raw archive.

- Feature engineering is correlation-based only; feature importance-driven selection and ablation studies remain future work [2].
- Full hyperparameter tuning and repeated cross-validation are not yet part of the pipeline.
- Concept drift evaluation and deployment-time latency profiling are deferred to Phase 2.
- The current Random Forest zero-day recall of 0.00 highlights the need for semi-supervised or transfer-learning extensions, as surveyed in [6].
- Evaluation against NSL-KDD [8] and CIC-IDS-2017 [7] is planned to enable cross-benchmark comparisons in Phase 2.

The next iteration should regenerate the processed artifact from the full raw data, persist model metrics to disk, and incorporate detection-threshold analysis for the anomaly component.

IX. Conclusion

This project delivers a coherent Phase-1 pipeline for adaptive cyber-physical intrusion detection on CIC-IDS-2018 [1]. It integrates literature-grounded motivation [2], [5], [6], dataset quality analysis, structured feature engineering, rigorous mathematical treatment of both models [3], [4], and empirical results evaluated through a dedicated zero-day recall metric. The key finding is that a supervised classifier alone cannot satisfy the open-world security objective: the Random Forest captures known attack patterns with high precision but fails entirely on unseen attack classes, while the One-Class SVM provides meaningful zero-day sensitivity at the cost of conservative recall—a trade-off consistently observed across benchmarks including NSL-KDD [8] and CIC-IDS-2017 [7]. Combining both paradigms is the recommended direction for adaptive security systems operating in dynamic threat environments [6].

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